

# Supplemental Material for “Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C<sub>60</sub> Graphullerene”

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## OVERVIEW

This Supplemental Material accompanies the main PRB article on coupled  $(\epsilon, \delta)$  response in qHP C<sub>60</sub>. All figures and tables are labeled with prefix **S** to match citations in the main text (Fig. S4–S6; Tables S1–S5). Calculations use the same PBE+DFT-D3 CP2K protocol as the main manuscript (rigid biaxial strain unless noted). Observables follow Capobianco *et al.* [1]:  $\pi$ -projected DOS, inverse participation ratio (IPR), nearest-neighbor coupling  $J$ , and Marcus reorganization energies  $\lambda^\pm$ . Converged inputs, outputs, and analysis JSON are archived as described in the main-text Data Availability statement.

## COMPUTATIONAL METHODS (FULL PROTOCOL)

All total-energy and electronic-structure calculations use CP2K [2] with PBE [3] and Grimme DFT-D3 (Becke–Johnson damping) [4]. Unless stated otherwise, biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol), providing an upper bound on strain–doping coupling until ionic-relaxation benchmarks in Table S3 converge. Plane-wave cutoff is 400 Ry for  $n \leq 4$  and 350 Ry for  $n \geq 6$  in archived Exp10 inputs (Table S2);  $\mathcal{S}(n)$  comparisons at fixed  $n$  use identical cutoffs.

Tetramer models ( $4 \times \text{C}_{60}$ , 240 sites) use B/N/P substitution at nominal 5% (12–13 dopants from archived coordinates, seed 42). Periodic scaling cells contain one dopant per C<sub>60</sub> ( $n = 1$ –8). Dopant sites are not re-optimized thermodynamically; fixed configurations probe mechanism rather than ensemble averages. A second random seed 137 tetramer grid (Table S4) bounds placement sensitivity.

Grids: 24 tetramer strain points ( $\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$ ); 12 tetramer electronic points; 40 periodic neutral-energy points at  $\epsilon \in \{0, +3\}\%$ ; charged-state optimizations for Marcus schematics (Fig. S5); IPR/ $J$  factorial (Fig. S6). Linear  $\alpha = dE/d\epsilon$  and substitution energies  $E_{\text{sub}}$  are defined in Table S1. **Task inventory (main-text support).** Mechanical strain: 24 tetramer points ( $\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$ , Table S1). Electronics: 12 tetramer PDOS/gap points ( $\epsilon \in \{-5, 0, +5\}\%$ , Fig. S4). Size scaling: 40 periodic neutral energies ( $n = 1$ –8,  $\epsilon \in \{0, +3\}\%$ ). Polaron fac-

torial: four neutral single-points (Fig. S6); charged-state GEO\_OPT and vertical SP for Fig. S5 (partial). SCF:  $10^{-6}$  Eh OT; basis DZVP-MOLOPT-SR-GTH.

**Non-additive coupling metric.** We quantify strain–doping cross terms with the synergy order parameter  $\mathcal{S}$ :

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (\text{S1})$$

where  $\epsilon$  is biaxial strain,  $\delta \in \{B, N, P\}$ , and  $\Delta E(\epsilon, \delta)$  is the energy per atom relative to the pristine, zero-strain reference at the same supercell size  $n \times \text{C}_{60}$ . The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives  $\mathcal{S} = 0$ ;  $\mathcal{S} < 0$  indicates cooperative stabilization beyond the two-scan sum, whereas  $\mathcal{S} > 0$  indicates cooperative destabilization.

To first order in strain,  $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon$  when  $\alpha_\delta = dE/d\epsilon$  is evaluated for the doped cell and  $\alpha_0$  for the pristine reference at the same  $n$  (rigid-strain protocol); this links the mechanical coefficient  $\alpha$  to the cross term at small  $\epsilon$ .

Periodic-cell  $\mathcal{S}$  values are obtained from archived total energies via post-processing of converged CP2K outputs (Supplemental Material Table S2). We report  $\mathcal{S}$  for total energy only; an analogous gap cross term  $\mathcal{S}_{\text{gap}}$  is not computed here—main-text panel (a) shows that gap slopes need not track  $|\mathcal{S}|$ .

**Localization and coupling (Supplemental Material Fig. S6).** Following Capobianco *et al.* [1], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling  $J$  from tetramer dimer fragments. IPR follows Supporting Information Eq. (S8),  $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$ , on a Mulliken-projected orbital grid;  $J$  is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant  $J \simeq 75$ –81 meV reported for pristine qHP networks [1], but they provide an internally consistent trend under coupled  $(\epsilon, \delta)$ .

**Polaron binding (Supplemental Material).** Marcus reorganization energies follow the vertical–adiabatic decomposition [1]:  $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ , with separate internal ( $\lambda_{\text{loc}}^\pm$ ) and environmental ( $\lambda_{\text{env}}^\pm$ ) contributions when geometry subsets are relaxed. Charged-state relaxations

TABLE S1. Total energies, linear strain sensitivities, and substitution energies for tetramer models (PBE+D3, rigid-strain protocol).  $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$  at  $\epsilon=0$  with no chemical-potential reference (not a standard formation enthalpy).  $n_{\text{dopants}}$  is counted from archived input coordinates (nominal 5% target; B: 13, N: 12, P: 12).

| System   | $n_{\text{dop}}$ | $E(0\%)$ (Ha) | $E(+5\%)$ (Ha) | $\alpha$ (meV/%) | $E_{\text{sub}}$ (eV/dopant) |
|----------|------------------|---------------|----------------|------------------|------------------------------|
| Pristine | —                | -1363.5512    | -1363.5509     | -1.0             | —                            |
| B-doped  | 13               | -1328.4900    | -1328.4090     | +56.9            | +73.4                        |
| N-doped  | 12               | -1414.3812    | -1414.4053     | -303.2           | -115.3                       |
| P-doped  | 12               | -1370.4965    | -1370.4888     | -19.1            | -15.8                        |

and vertical single-points are documented in Supplemental Material Fig. S5.

TABLE S1

TABLE S2

TABLE S4

TABLE S5

TABLE S3

FIGURE S4

FIGURE S5

FIGURE S6

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- [1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, Electron localization and mobility in monolayer fullerene networks, *Nano Letters* **24**, 7891 (2024).
  - [2] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, Quickstep: Fast and accurate density functional calculations using a mixed gaussian and plane waves approach, *Computer Physics Communications* **167**, 103 (2005).
  - [3] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Physical Review Letters* **77**, 3865 (1996).
  - [4] S. Grimme, S. Ehrlich, and L. Goerigk, Effect of the damping function in dispersion corrected density functional theory, *Journal of Computational Chemistry* **32**, 1456 (2011).

TABLE S2. Summary of converged DFT calculations and correspondence to main-text results. The  $\sim 90$  tasks cited in the main text are the converged calculations that directly support quantitative claims; the full archive includes additional preliminary grids (rows marked “not used”).

| Calculation set             | Model                    | Converged tasks | Main-text usage       | Notes                                        |
|-----------------------------|--------------------------|-----------------|-----------------------|----------------------------------------------|
| Early strain probe          | $2 \times C_{60}$        | 5               | not used              | superseded by tetramer grids                 |
| Single-molecule doping      | $C_{60}$                 | 11              | not used              | superseded                                   |
| Early electronics           | $2 \times C_{60}$        | 24              | not used              | superseded by tetramer PDOS                  |
| Polaron IPR/ $J$ factorial  | tetramer                 | 4               | Fig. S6; main inset   | pristine/ $B \times \epsilon=0, +3\%$        |
| Tetramer strain energies    | tetramer                 | 24              | Table S1, $\alpha$    | rigid strain                                 |
| Optimal-design grid         | tetramer                 | 20              | not used              | archive only                                 |
| Tetramer PDOS/gaps          | tetramer                 | 12              | Fig. S4; main panels  |                                              |
| Geometry references         | tetramer relax           | 6               | charged-state inputs  | reference geometries                         |
| Charged polarons            | tetramer $q = \pm 1$     | partial         | Fig. S5               | $\lambda$ pending vertical SP                |
| Size scaling                | $n \times C_{60}$        | 40              | main $\mathcal{S}(n)$ | 400 Ry ( $n \leq 4$ ), 350 Ry ( $n \geq 6$ ) |
| $n=6$ cutoff control        | $6 \times C_{60}$ N @+3% | pending         | N sign benchmark      | 400 Ry ( <b>cutoff400</b> inp)               |
| Ionic relaxation (tetramer) | tetramer                 | pending         | Table S3              | fixed-cell GEO_OPT                           |

TABLE S3. Alternate random-seed dopant placement (tetramer, rigid strain, PBE+D3). Seed 42 is used in the main text; seed 137 repeats the full B/N/P strain grid ( $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ ) to test whether  $\alpha$  and tetramer  $\mathcal{S}$  at +3% are placement-sensitive. Periodic  $n=4$   $\mathcal{S}$  values (seed 42) are listed for comparison only—different concentration and boundary conditions.

| Dopant | Seed | $\alpha$ (meV/%) | $\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom) | $\mathcal{S}_{+3\%}^{n=4}$ (meV/atom) | Status              |
|--------|------|------------------|----------------------------------------------|---------------------------------------|---------------------|
| B      | 42   | +56.9            | —                                            | −6.2                                  | verified (Table S1) |
| N      | 42   | −303.2           | —                                            | +3.0                                  | verified            |
| P      | 42   | −19.1            | —                                            | −23.7                                 | verified            |
| B      | 137  | pending          | pending                                      | —                                     | 6 inp / pending     |
| N      | 137  | pending          | pending                                      | —                                     | 6 inp / pending     |
| P      | 137  | pending          | pending                                      | —                                     | 6 inp / pending     |

TABLE S4. Local dopant–cage structure from archived tetramer XYZ (mean nearest C distance  $\bar{d}$  in Å; covalent-radius excess vs. C in pm). Strain-induced change  $\Delta\bar{d} = \bar{d}(\epsilon=+3\%) - \bar{d}(\epsilon=0)$  summarizes how tensile load couples to pre-existing distortion (see Discussion mechanistic synthesis).

| Dopant | $\bar{d}(\epsilon=0)$ (Å) | $\bar{d}(\epsilon=+3\%)$ (Å) | $\Delta\bar{d}$ (Å) | $\Delta r_{\text{cov}}$ (pm) |
|--------|---------------------------|------------------------------|---------------------|------------------------------|
| B      | 1.384                     | 1.393                        | +0.0085             | +7                           |
| N      | 1.393                     | 1.402                        | +0.0087             | −6                           |
| P      | 1.384                     | 1.385                        | +0.00003            | +30                          |

TABLE S5. Ionic-relaxation benchmark for the tetramer  $P@+3\%$  synergy corner (four fixed-cell GEO\_OPT corners). Main-text  $\mathcal{S}(n)$  in panel (d) of the composite figure uses *periodic*  $n \times C_{60}$  rigid-strain energies (PBE+D3); this table tests whether a tetramer cross term survives atomic relaxation under the same biaxial cell. Rigid tetramer energies are from the archived tetramer strain-energy grid (PBE, no D3); relaxed energies use PBE+D3 fixed-cell GEO\_OPT (300 Ry) as described in the Data Availability statement.  $\mathcal{S}$  follows Eq. (S1) with  $N=240$  atoms.

| Quantity                                                 | Rigid (legacy tetramer)   | Relaxed (GEO_OPT) |
|----------------------------------------------------------|---------------------------|-------------------|
| $E(\text{pristine}, \epsilon=0)$ (Ha)                    | −1363.5512                | pending           |
| $E(\text{pristine}, \epsilon=3\%)$ (Ha)                  | −1363.5522                | pending           |
| $E(P, \epsilon=0)$ (Ha)                                  | −1370.4965                | pending           |
| $E(P, \epsilon=3\%)$ (Ha)                                | −1370.4890                | pending           |
| $\mathcal{S}$ (meV/atom)                                 | +0.96                     | pending           |
| Sign vs. periodic $n=4$ $\mathcal{S}_P$ (−23.7 meV/atom) | compare after convergence |                   |

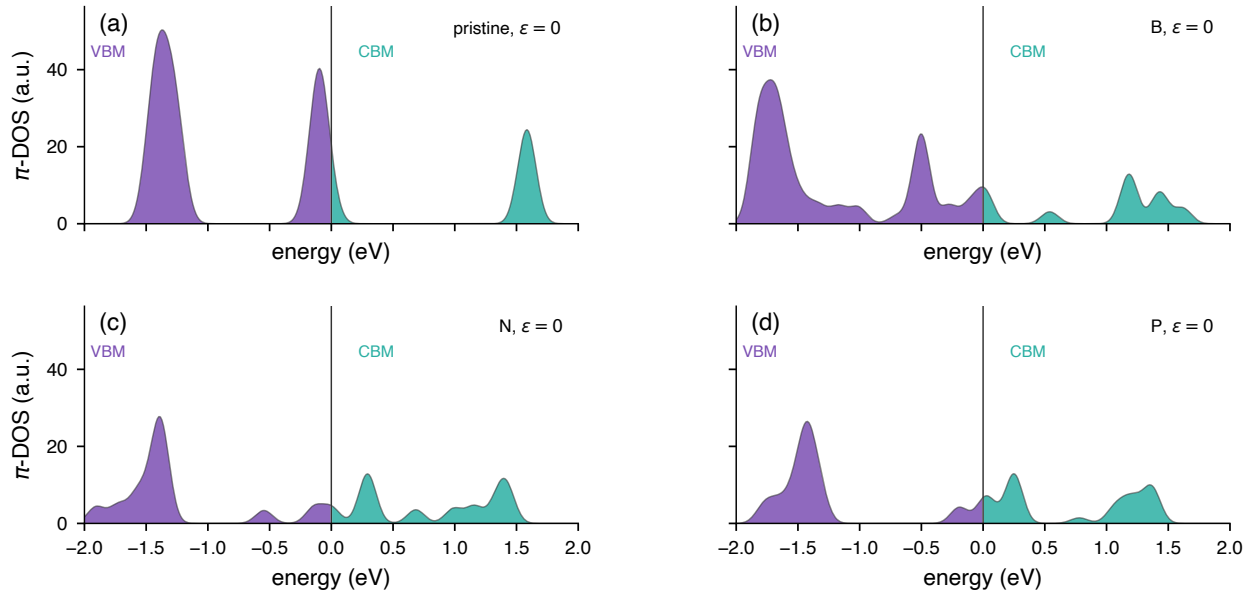


FIG. S1.  $\pi$ -projected density of states ( $\pi$ -DOS) for pristine and B/N/P-doped tetramers at  $\epsilon=0$  (PBE+D3). Panels (a)–(d) share a common vertical scale for cross-dopant comparison. Valence band edges (purple) and conduction band edges (teal) follow the color scheme of Capobianco *et al.* [1]. Energy is relative to the Fermi level at 0 eV. Molecular-orbital isosurfaces are pending cube export.

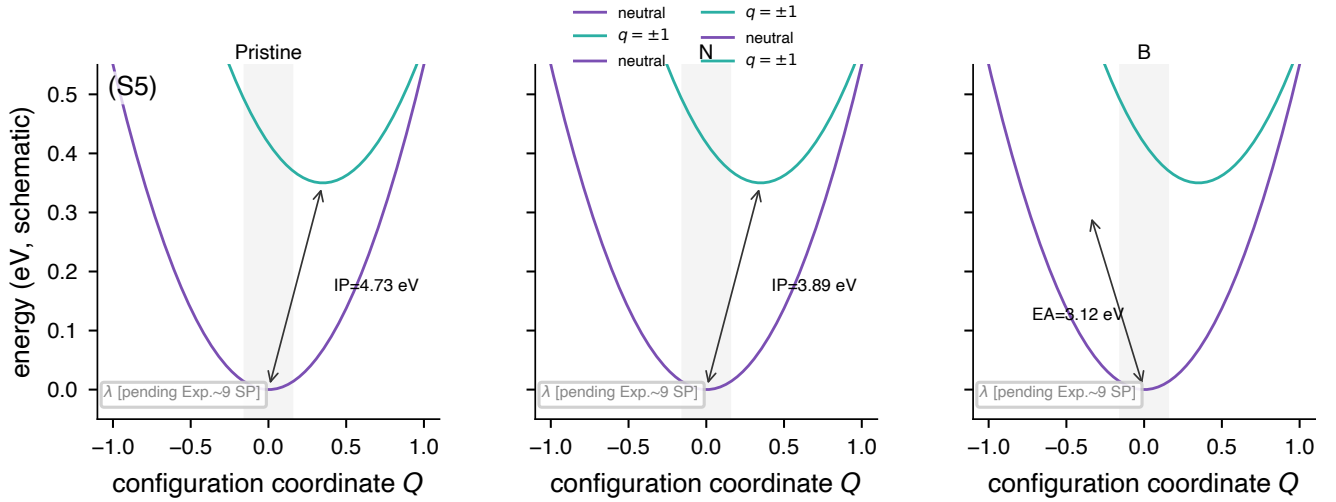


FIG. S2. Schematic configuration-coordinate diagrams for Marcus reorganization energies, following Ref. 1. Panels: pristine (hole,  $\lambda^+$ ), N-doped (hole), and B-doped (electron,  $\lambda^-$ ). Adiabatic ionization potential (IP) and electron affinity (EA) are from converged charged-state geometry optimizations;  $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$  remains pending vertical single-points at the neutral geometry.

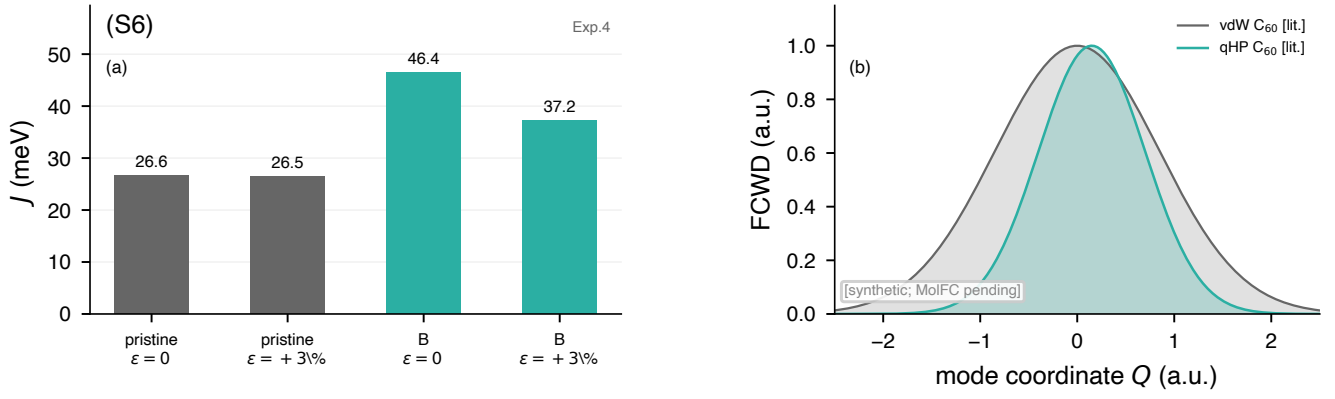


FIG. S3. Transport-adjacent observables (PBE+D3). **(a)** Nearest-neighbor electronic coupling  $J$  on the  $2 \times 2$  factorial (pristine/B  $\times \epsilon=0, +3\%$ ); single-point values without statistical error bars. **(b)** Illustrative Franck-Condon weighted density (FCWD) envelope—a *schematic placeholder* only; tetramer MolFC generating-function calculations are pending and this panel must not be read as a computed spectrum. The horizontal axis is a dimensionless mode coordinate  $Q$ .